

Notes on the Agda Development for Qupit Clifford Presentations

June 23, 2026

1 Overview

This repository contains the Agda development accompanying the paper “A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions.” The formalization is organized around presentations of groups by generators and relations. Words are built from gates such as S , H , and CZ ; relations are encoded as Agda constructors; and derivations are proofs in the congruence closure generated by those constructors.

There are two intertwined pieces of structure. First, the symplectic normal-form proof reduces the large family of box relations to a small set of axioms. Second, the Clifford presentation modulo scalars is obtained from a semidirect product of Pauli translations and symplectic gates. The semidirect product is not only a conceptual explanation: it is used in the code to transport relations from the symplectic and Pauli presentations into the Clifford presentation.

The most relevant files are:

<code>N/Boxes.agda</code> , <code>N/Boxes-Sym.agda</code>	box-shaped normal-form indices,
<code>N/BR/One</code> , <code>N/BR/Two</code> , <code>N/BR/Three</code>	box-relation derivations,
<code>N/Symplectic-Simplified.agda</code>	small symplectic relation set,
<code>N/Clifford/SDProduct.agda</code>	Pauli–symplectic semidirect product,
<code>N/Clifford/Iso.agda</code> , <code>N/Clifford/Iso3.agda</code>	isomorphism with the Clifford presentation,
<code>N/Clifford/Clifford-Mod-Scalar.agda</code>	Clifford modulo scalars.

2 How Box Relations Are Reduced

2.1 Boxes and normal forms

The box language packages repeated circuit fragments into finite families of parameters over $\mathbb{Z}_p$. In `N/Boxes.agda` and `N/Boxes-Sym.agda`, the main families are:

$$\begin{aligned} A &= \{(a, b) \in \mathbb{Z}_p^2 \mid (a, b) \neq (0, 0)\}, \\ B &= \mathbb{Z}_p^2, \\ D &= \mathbb{Z}_p^2, \\ E &= \mathbb{Z}_p. \end{aligned}$$

The types L , M , LM , and NF assemble these boxes into width-indexed normal forms. Informally, an $NF(n)$ is a triangular normal form for an n -qupit circuit: it consists of an $NF(n-1)$ part followed by the last layer $LM(n)$. The proof task is then: when a generator is multiplied against a box normal form, rewrite the product back into the appropriate box normal form.

The reduction of box relations is deliberately mechanical. A typical proof has the following shape.

1. Expand the named box into the underlying gate word.

2. Use local presentation relations and already-derived pushing lemmas to move the incoming generator through the box.
3. Collect the remaining word back into a box constructor.
4. Discharge the parameter update by arithmetic in $\mathbb{Z}_p$.

The box-relation files are split by arity and by box family. For example, `N/BR/One/A.agda` proves one-qupit A -box relations, while `N/BR/Two/L-CZ.agda` proves the rules for pushing a CZ through a two-qupit L box. The arithmetic side conditions used by several proofs are collected in `N/BR/Calculations.agda`.

2.2 A simple reduction example

One of the smallest examples is the one-qupit E -box relation shown in the existing literate documentation. The E box is represented by a single parameter $b \in \mathbb{Z}_p$, and its interpretation is

$$[b]^e = S^{-b}.$$

Multiplying by S on the right gives:

$$\begin{aligned} [b]^e S &= S^{-b} S \\ &= S^{-b+1} \\ &= S^{-(b-1)} \\ &= [b-1]^e. \end{aligned}$$

In Agda this is the proof `lemma-single-qupit-br-E` in `doc/Readme.lagda.tex`. The only group-theoretic content is the power-combination lemma for S ; the rest is modular arithmetic, namely rewriting $-b+1$ as $-(b-1)$ in $\mathbb{Z}_p$.

This example illustrates the pleasant case: the shape of the box does not change, and the whole relation is a parameter update.

2.3 A more complicated reduction example

A more representative example is the general nonzero A -box pushed through H . For $a, b \in \mathbb{Z}_p^\times$, the relation proved in `N/BR/One/A.agda` is:

$$[(a, b)]^a H \approx S^{(ab)^{-1}} [(b, -a)]^a.$$

This is the Agda theorem `fig-24-3`. The proof begins by expanding the A box into the one-qupit normal-form word used by the development. After expansion, the core subword is of the form

$$M(a^{-1}) H S^{-b/a} H.$$

The derived pushing lemma `derived-7` rewrites this into a word whose parameters are not yet in the desired box format:

$$S^{-x^{-1}y^2} M(-y/x) H S^{-x^{-1}}, \quad x = -b/a, \quad y = a^{-1}.$$

The remaining work is arithmetic. The file `N/BR/Calculations.agda` proves exactly the identities needed to identify these parameters with the target box:

$$-x^{-1}y^2 = (ab)^{-1}, \quad -y/x = b^{-1}, \quad -x^{-1} = ab^{-1}.$$

After these substitutions, the word folds back into

$$S^{(ab)^{-1}} [(b, -a)]^a.$$

This example shows why the box-relation layer is useful. The high-level relation is simple to state, but its proof combines presentation rewriting, normal-form recognition, nonzero side conditions, and modular-field calculations. Agda keeps all of those bookkeeping steps explicit.

3 From Symplectic and Pauli Relations to Clifford Relations

3.1 The semidirect product presentation

The Clifford group modulo scalars decomposes as a Pauli translation part acted on by a symplectic part. In the code this is made precise in `N/Clifford/SDProduct.agda`. The generator type is a tagged sum:

$$\text{Gen}_{SD}(n) = \text{Gen}_{XZ}(n) \sqcup \text{Gen}_{Sym}(n).$$

The left side contains the Pauli generators X and Z ; the right side contains the symplectic gates H , S , and CZ . The relation set is built from three pieces:

$$\Gamma_{SD} = \Gamma_{XZ} \cup \Gamma_{Sym} \cup \Gamma_{\text{conj}}.$$

The first two pieces are the Pauli and symplectic presentations. The third piece records the action of the symplectic generators on Pauli generators. For example, `conj` in `SDProduct.agda` includes:

$$\begin{array}{ll} HXH^{-1} &= Z, & HZH^{-1} &= X^{-1}, \\ SX S^{-1} &= XZ, & SZ S^{-1} &= Z, \\ CZ X_{\text{top}} CZ^{-1} &= X_{\text{top}} Z_{\text{bottom}}, & CZ X_{\text{bottom}} CZ^{-1} &= X_{\text{bottom}} Z_{\text{top}}. \end{array}$$

These are the familiar Clifford conjugation rules. In the semidirect product, they are not derived later as accidental lemmas; they are the middle relations that say how the right-hand symplectic factor acts on the left-hand Pauli factor.

3.2 Embedding the symplectic presentation into Clifford

The important point is that the symplectic generator S does not embed as the raw Clifford generator named S . In `N/Clifford/Clifford-Mod-Scalar.agda`, the Clifford presentation defines derived Pauli words

$$Z = HSHHS^{-1}, \quad X = HSHHS^{-1}H,$$

and then defines the symplectic shear

$$s = SZ^{1/2}.$$

This derived word s is the image of the symplectic S in the Clifford presentation.

The forward map from the semidirect product presentation to the Clifford presentation is defined in `N/Clifford/Iso.agda`:

$$\begin{array}{ll} f(X) &= \text{derived Clifford } X, \\ f(Z) &= \text{derived Clifford } Z, \\ f(H) &= H, \\ f(S_{\text{sym}}) &= s = S_{\text{cliff}} Z^{1/2}, \\ f(CZ) &= CZ. \end{array}$$

This is the embedding of the symplectic presentation into Clifford: H and CZ embed directly, while the symplectic shear embeds as the Clifford phase gate corrected by a half- Z Pauli. That

correction is exactly what makes the symplectic action match the Clifford action modulo scalars in odd prime dimension.

The inverse map explains the correction from the other direction:

$$\begin{aligned} h(H_{\text{cliff}}) &= H, \\ h(S_{\text{cliff}}) &= Z^{-1/2} S_{\text{sym}}, \\ h(CZ_{\text{cliff}}) &= CZ. \end{aligned}$$

The proof `f-left-inv-gen` in `N/Clifford/Iso3.agda` checks that these two descriptions cancel, using $Z^{-1/2}Z^{1/2} = 1$ and the commutation of Z with S in the relevant one-qupit fragment.

3.3 Deriving Clifford relations

Once the embedding is fixed, the Clifford relations modulo scalars are transported from the semidirect product. The symplectic relations become Clifford relations after replacing

$$S_{\text{sym}} \quad \text{by} \quad s = S_{\text{cliff}} Z^{1/2}.$$

For example, the symplectic diagonal relation

$$Mg S = S^{g^2} Mg$$

is transported to the Clifford relation

$$Mg s = s^{g^2} Mg,$$

which appears as `semi-Ms` in `N/Clifford/Clifford-Mod-Scalar.agda`. Similarly, the symplectic M word

$$M(x) = S^x H S^{x^{-1}} H S^x H$$

is sent to the Clifford word with every symplectic S replaced by s .

The Pauli relations and the mixed action relations give the remaining Clifford–Pauli transport rules. For instance, the semidirect action of CZ on X becomes:

$$CZ X_{\text{top}} = X_{\text{top}} Z_{\text{bottom}} CZ, \quad CZ X_{\text{bottom}} = X_{\text{bottom}} Z_{\text{top}} CZ,$$

which are the relations `rel-X↑-CZ` and `rel-X↓-CZ` in the Clifford presentation.

The isomorphism theorem in `N/Clifford/Iso3.agda` packages this transport formally:

$$\mathcal{P} \rtimes \mathcal{S} \cong \mathcal{C}/\text{scalars}.$$

Its proof supplies both directions: every semidirect-product relation is sound in the Clifford presentation, and every Clifford generator is represented by a semidirect-product word.

4 Why the Simplified Clifford Relations Have Pauli Tails

The file `N/Clifford/Clifford-Mod-Scalars-Simplified.agda` tries to state the Clifford relations using the raw gate S as much as possible, rather than the symplectic shear $s = SZ^{1/2}$. Most relations are unchanged. The interesting cases are the two Selinger $CZHCZ$ relations.

In the purely symplectic presentation, one has for example:

$$CZ H_{\text{top}} CZ = S_{\text{top}}^{-1} H_{\text{top}} S_{\text{top}}^{-1} CZ H_{\text{top}} S_{\text{top}}^{-1} S_{\text{bottom}}^{-1}.$$

In the Clifford presentation, however, the embedded symplectic S^{-1} is s^{-1} , not the raw Clifford S^{-1} . Replacing s^{-1} by raw S^{-1} introduces half- Z corrections. These corrections are pushed through the surrounding H and CZ gates using the Pauli action rules. The corrections then cancel except for a rightmost Pauli tail. The simplified relation therefore becomes:

$$CZ H_{\text{top}} CZ X_{\text{top}} Z_{\text{top}} = S_{\text{top}}^{-1} H_{\text{top}} S_{\text{top}}^{-1} CZ H_{\text{top}} S_{\text{top}}^{-1} S_{\text{bottom}}^{-1}.$$

This is `selinger-c10` in the simplified Clifford file; the bottom-qubit analogue is `selinger-c11`. Their soundness is verified in `N/Clifford/Clifford-Simplified-Verify.agda`, and the identity isomorphism between the original and simplified Clifford presentations is recorded in `N/Clifford/Iso-Clifford-Simplified.agda`.